Blockchain Lab
Experiment 9

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Aim: To implement private Blockchain using Geth.

Theory:

A **blockchain** is a distributed and decentralized ledger that records transactions securely across multiple nodes.

Geth (Go Ethereum) is the official command-line client used to run an Ethereum node. It allows you to:

- Create a private blockchain
- Mine blocks
- Deploy smart contracts
- Perform transactions

Private Blockchain

A **private blockchain** is a restricted network where:

- Only authorized participants can join
- It is controlled by a single organization or group
- Faster and more efficient than public blockchains

Key Concepts

- **Genesis Block**
  The first block of the blockchain. It defines:
  - Chain configuration
  - Initial accounts
  - Difficulty level

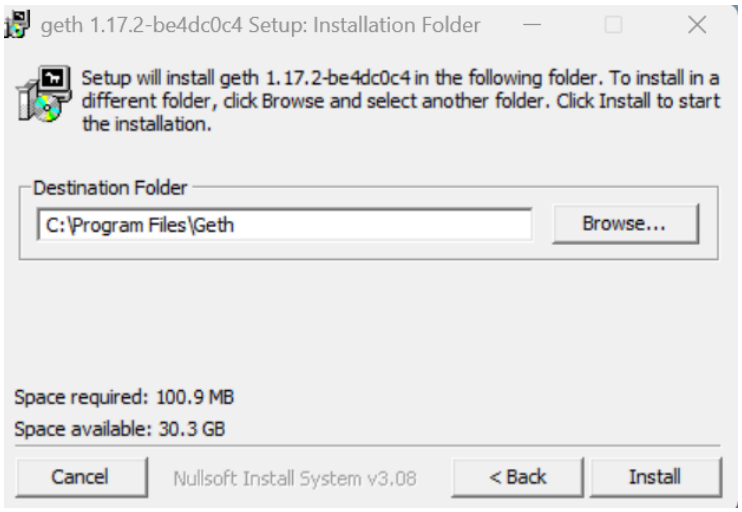
- **Node**
  A system running Geth that participates in the blockchain network.
- **Mining**
  The process of validating transactions and adding blocks to the chain.
- **Account**
  Used to send/receive Ether in the network.

Steps to Implement Private Blockchain using Geth

1. Install Geth on your system (Windows/Linux).
2. Create a working directory for your private blockchain.
3. Create a genesis file to define the blockchain configuration (chain ID, difficulty, gas limit, etc.).
4. Initialize the blockchain using the genesis file.
5. Create a new account to use in the blockchain.
6. Start the Geth node with a custom network ID.
7. Open the Geth console to interact with the blockchain.
8. Start mining to generate blocks and Ether.
9. Check account balance to verify mining rewards.
10. Perform transactions between accounts.
11. View blockchain details like block number and block data.

Implementation:

Install Geth for windows from Ethereum website



Verify in command prompt if Geth is installed by using geth version

```
C:\Users\HOME\spandan>geth version
Geth
Version: 1.17.2-stable
Git Commit: be4dc0c4be2fe316dbdd0a73e48421f64978232f
Git Commit Date: 20260330
Architecture: amd64
Go Version: go1.25.7
Operating System: windows
GOPATH=
GOROOT=
```

Create project folder

```
C:\Users\HOME\spandan>mkdir private_blockchain

C:\Users\HOME\spandan>cd private_blockchain

C:\Users\HOME\spandan\private_blockchain>
```

Create a genesis.json file in notepad

```
C:\Users\HOME\spandan\private_blockchain>notepad genesis.json
```

```
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "istanbulBlock": 0
  },
  "difficulty": "2000",
  "gasLimit": "2100000",
  "alloc": {}
}
```

Initialize the private blockchain using geth

```
C:\Users\HOME\spandan\private_blockchain>geth --datadir data init genesis.json
INFO [04-07|15:09:44.624] Maximum peer count                       ETH=50 total=50
WARN [04-07|15:09:44.640] Sanitizing cache to Go's GC limits       provided=4096 updated=2018
INFO [04-07|15:09:44.642] Set global gas cap                       cap=50,000,000
INFO [04-07|15:09:44.642] Initializing the KZG library             backend=gokzg
INFO [04-07|15:09:44.650] Using pebble as the backing database
INFO [04-07|15:09:44.650] Allocated cache and file handles         database=C:\Users\HOME\spandan\private_blockchain\data\geth\chaind
ata cache=1009.00MiB handles=8192
INFO [04-07|15:09:44.751] Opened ancient database                  database=C:\Users\HOME\spandan\private_blockchain\data\geth\chaind
ata\ancient\chain readonly=false
INFO [04-07|15:09:44.752] Opened Era store                         datadir=C:\Users\HOME\spandan\private_blockchain\data\geth\chainda
ta\ancient\chain\era
INFO [04-07|15:09:44.752] State schema set to default              scheme=path
WARN [04-07|15:09:44.752] Head block is not reachable
INFO [04-07|15:09:44.752] Load database journal from disk
INFO [04-07|15:09:44.760] Opened ancient database                  database=C:\Users\HOME\spandan\private_blockchain\data\geth\chaind
ata\ancient\state readonly=false
INFO [04-07|15:09:44.760] State snapshot generator is not found
INFO [04-07|15:09:44.761] Starting snapshot generation             root=56e81f..63b421 accounts=0 slots=0 storage=0.00B dangling=0 el
apsed="545.4µs"
INFO [04-07|15:09:44.761] Initialized path database                triecache=16.00MiB statecache=16.00MiB buffer=64.00MiB state-histo
ry="last 90000 blocks" journal-dir=C:\Users\HOME\spandan\private_blockchain\data\geth\triedb
INFO [04-07|15:09:44.761] Resuming snapshot generation             root=56e81f..63b421 accounts=0 slots=0 storage=0.00B dangling=0 el
apsed="545.4µs"
INFO [04-07|15:09:44.761] Writing custom genesis block
INFO [04-07|15:09:44.761] Generated snapshot                       accounts=0 slots=0 storage=0.00B dangling=0 elapsed=1.154ms
INFO [04-07|15:09:44.767] Successfully wrote genesis state         database=chaindata hash=6f14b1..a2805e
```

Create an account

```
C:\Users\HOME\spandan\private_blockchain>geth --datadir data account new
INFO [04-07|15:12:54.245] Maximum peer count                       ETH=50 total=50
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x0C0F4Cd470564897545A41A3C8243e60430ce94B
Path of the secret key file: data\keystore\UTC--2026-04-07T09-43-19.390875900Z--0c0f4cd470564897545a41a3c8243e60430ce94b

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Start Geth Node

```
C:\Users\HOME\spandan\private_blockchain>geth --datadir data --networkid 1234 console
INFO [04-07|16:50:01.532] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-07|16:50:01.595] Set global gas cap                       cap=25000000
INFO [04-07|16:50:01.603] Allocated trie memory caches             clean=256.00MiB dirty=256.00MiB
INFO [04-07|16:50:01.611] Allocated cache and file handles         database=C:\Users\HOME\spandan\private_blockchain\data\geth\chaind
ata cache=512.00MiB handles=8192
INFO [04-07|16:50:01.665] Opened ancient database                  database=C:\Users\HOME\spandan\private_blockchain\data\geth\chaind
ata\ancient
INFO [04-07|16:50:03.008] Initialised chain configuration          config="{ChainID: 1234 Homestead: 0 DAO: <nil> DAOSupport: false E
IP150: 0 EIP155: 0 EIP158: 0 Byzantium: 0 Constantinople: 0 Petersburg: 0 Istanbul: 0, Muir Glacier: <nil>, YOLO v2: <nil>, Engine: u
nknown}"
INFO [04-07|16:50:03.034] Disk storage enabled for ethash caches   dir=C:\Users\HOME\spandan\private_blockchain\data\geth\ethash coun
t=3
INFO [04-07|16:50:03.046] Disk storage enabled for ethash DAGs     dir=C:\Users\HOME\AppData\Local\Ethash count=2
INFO [04-07|16:50:03.056] Initialising Ethereum protocol           versions="[65 64 63]" network=1234 dbversion=<nil>
WARN [04-07|16:50:03.067] Upgrade blockchain database version      from=<nil> to=8
INFO [04-07|16:50:03.077] Loaded most recent local header          number=0 hash="6f14b1…a2805e" td=2000 age=57y1mo11h
INFO [04-07|16:50:03.091] Loaded most recent local full block      number=0 hash="6f14b1…a2805e" td=2000 age=57y1mo11h
INFO [04-07|16:50:03.104] Loaded most recent local fast block      number=0 hash="6f14b1…a2805e" td=2000 age=57y1mo11h
INFO [04-07|16:50:03.119] Regenerated local transaction journal    transactions=0 accounts=0
INFO [04-07|16:50:03.141] Allocated fast sync bloom                size=512.00MiB
INFO [04-07|16:50:03.148] Initialized fast sync bloom              items=0 errorrate=0.000 elapsed=0s
INFO [04-07|16:50:03.149] Starting peer-to-peer node               instance=Geth/v1.9.25-stable-e7872729/windows-amd64/go1.15.6
INFO [04-07|16:50:03.190] New local node record                    seq=1 id=eeeee1fa666f7c09 ip=127.0.0.1 udp=30303 tcp=30303
INFO [04-07|16:50:03.203] Started P2P networking                   self=enode://a3e0ec411b9c6c658245a289eb62ae41f93a67c9839cf01d06840
f25ea35bd73417385b2fe1b7c7fc18b1f111b097e6ac0981ace7aee219532c88296e496b93b@127.0.0.1:30303
INFO [04-07|16:50:03.206] IPC endpoint opened                      url=\\.\pipe\geth.ipc
INFO [04-07|16:50:03.319] Etherbase automatically configured       address=0xa0c392e16b0F0e576683d2e5d51FaFAd2AcA6e4f
Welcome to the Geth JavaScript console!
```

Check accounts

```
> eth.accounts
["0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f"]
```

Unlock the account

```
> personal.unlockAccount(eth.accounts[0])
Unlock account 0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f
Passphrase: INFO [04-07|16:56:23.593] Looking for peers                peercount=0 tried=66 static=0

true
```

Start Mining

```
> miner.start(1)
INFO [04-07|16:57:49.057] Updated mining threads                   threads=1
INFO [04-07|16:57:49.063] Transaction pool price threshold updated price=1000000000
null
> INFO [04-07|16:57:49.070] Commit new mining work                   number=1 sealhash="cb5512…b7dc49" uncles=0 txs=0 gas=0 fees=0 el
apsed=1.007ms
INFO [04-07|16:57:50.937] Generating DAG in progress               epoch=0 percentage=0 elapsed=1.106s
INFO [04-07|16:57:52.040] Generating DAG in progress               epoch=0 percentage=1 elapsed=2.210s
INFO [04-07|16:57:53.133] Generating DAG in progress               epoch=0 percentage=2 elapsed=3.302s
INFO [04-07|16:57:54.226] Generating DAG in progress               epoch=0 percentage=3 elapsed=4.396s
INFO [04-07|16:57:55.338] Generating DAG in progress               epoch=0 percentage=4 elapsed=5.507s
INFO [04-07|16:57:55.981] Looking for peers                        peercount=0 tried=59  static=0
INFO [04-07|16:57:56.461] Generating DAG in progress               epoch=0 percentage=5 elapsed=6.631s
INFO [04-07|16:57:57.583] Generating DAG in progress               epoch=0 percentage=6 elapsed=7.753s
INFO [04-07|16:57:58.706] Generating DAG in progress               epoch=0 percentage=7 elapsed=8.876s
INFO [04-07|16:57:59.803] Generating DAG in progress               epoch=0 percentage=8 elapsed=9.973s
```

```
INFO [04-07|20:45:12.391] Commit new mining work                   number=2 sealhash="4f33f6…a9b2e0" uncles=0 txs=0 gas=0 fees=0 elap
sed="998µs"
INFO [04-07|20:45:12.401] 🔨 mined potential block                  number=1 hash="e781cd…21e1bb"
INFO [04-07|20:45:12.916] Successfully sealed new block            number=2 sealhash="4f33f6…a9b2e0" hash="5590c8…45f96b" elapsed=524
.571ms
INFO [04-07|20:45:12.916] Commit new mining work                   number=3 sealhash="2fe669…f8004b" uncles=0 txs=0 gas=0 fees=0 elap
sed=0s
INFO [04-07|20:45:12.926] 🔨 mined potential block                  number=2 hash="5590c8…45f96b"
INFO [04-07|20:45:15.455] Successfully sealed new block            number=3 sealhash="2fe669…f8004b" hash="c5becd…86cb9c" elapsed=2.5
39s
INFO [04-07|20:45:15.455] Commit new mining work                   number=4 sealhash="b6f543…2d8eb5" uncles=0 txs=0 gas=0 fees=0 elap
sed=0s
INFO [04-07|20:45:15.466] 🔨 mined potential block                  number=3 hash="c5becd…86cb9c"
INFO [04-07|20:45:17.874] Successfully sealed new block            number=4 sealhash="b6f543…2d8eb5" hash="4bbde3…c3cd16" elapsed=2.4
19s
```

Check account balance

```
> eth.getBalance(eth.accounts[0])
2800000000000000000
> INFO [04-07|20:57:20.259] Looking for peers
```

Stop mining

```
> miner.stop()
null
> INFO [04-07|20:57:52.482] Looking for peers                        peercount=0 tried=43 static=0
INFO [04-07|20:58:02.489] Looking for peers                        peercount=0 tried=50 static=0
```

To Send transaction

Created another account

```
C:\Users\HOME\spandan\private_blockchain>geth --datadir data account new
INFO [04-07|21:07:14.350] Maximum peer count                       ETH=50 LES=0 total=50
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0xc5Ab7B0f31c6304828eeC594b9eA37Af6D07f845
Path of the secret key file: data\keystore\UTC--2026-04-07T15-37-22.772205900Z--c5ab7b0f31c6304828eec594b9ea37af6d07f845

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

Now we have 2 accounts

```
> eth.accounts
["0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f", "0xc5ab7b0f31c6304828eec594b9ea37af6d07f845"]
> INFO [04-07|21:09:10.220] Looking for peers                        peercount=0 tried=22 static=0
```

Unlock the account 0 and send transaction to account 1

```
> personal.unlockAccount(eth.accounts[0])
Unlock account 0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f
Passphrase: INFO [04-07|21:16:11.421] Looking for peers                    peercount=0 tried=57 static=0

true
> INFO [04-07|21:16:22.171] Looking for peers                    peercount=0 tried=40 static=0
> eth.sendTransaction({
......   from: eth.accounts[0],
......   to: eth.accounts[1],
......   value: web3.toWei(1, "ether")
...... })
INFO [04-07|21:16:31.327] Setting new local account                address=0xa0c392e16b0F0e576683d2e5d51FaFAd2AcA6e4f
INFO [04-07|21:16:31.337] Submitted transaction                    fullhash=0x61629c85ab48b216143d2e288f5c6c30198fd6e81416315c516dcbd
5d7fa6982 recipient=0xc5Ab7B0f31c6304828eeC594b9eA37Af6D07f845
"0x61629c85ab48b216143d2e288f5c6c30198fd6e81416315c516dcbd5d7fa6982"
> INFO [04-07|21:16:32.176] Looking for peers                    peercount=0 tried=21 static=0
INFO [04-07|21:16:42.293] Looking for peers                    peercount=0 tried=37 static=0
INFO [04-07|21:16:52.437] Looking for peers                    peercount=0 tried=43 static=0
INFO [04-07|21:17:02.442] Looking for peers                    peercount=0 tried=22 static=0
```

Start mining

```
> miner.start(1)
INFO [04-07|21:19:03.349] Updated mining threads                   threads=1
INFO [04-07|21:19:03.355] Transaction pool price threshold updated price=1000000000
INFO [04-07|21:19:03.361null] Com
mit new>  mining work                  number=23 sealhash="dcf750…5b31fd" uncles=0 txs=0 gas=0 fees=0 elapsed=0s
INFO [04-07|21:19:03.375] Commit new mining work                   number=23 sealhash="960b1a…081554" uncles=0 txs=1 gas=21000 fees=2
.1e-05 elapsed=13.991ms
INFO [04-07|21:19:03.790] Looking for peers                    peercount=0 tried=32 static=0
INFO [04-07|21:19:14.580] Looking for peers                    peercount=0 tried=36 static=0
```

Stop mining

```
> miner.stop()
null
> INFO [04-07|21:21:36.229] Looking for peers
```

Account Balances

```
> eth.getBalance(eth.accounts[0])
113000000000000000000
> INFO [04-07|21:24:18.456] Looking for peers                    peercount=0 tried=45 static=0
> eth.getBalance(eth.accounts[1])
1000000000000000000
```

Transaction was successful as balance of account 0 has decreased and balance of account 1 is 1 ether which was transferred to account 1.

To fetch the details of the latest block mined

```
> eth.getBlock("latest")
{
  difficulty: 133185,
  extraData: "0xda830109198467657468886 76f312e31352e368777696e646f7773",
  gasLimit: 2220057,
  gasUsed: 0,
  hash: "0xa2f12559d9f4bca9134f88ac1485816862f83f12e4e29d4a168bf0de50bf2cc6",
  logsBloom: "0x0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000",
  miner: "0xa0c392e16b0f0e576683d2e5d51fafad2aca6e4f",
  mixHash: "0x4f3da61e221de9d1a093c054331cfcd2ab1d31e2ad0fd6e994750003a715058e",
  nonce: "0x0721823492469cc9",
  number: 57,
  parentHash: "0x248ed1917d5a5ef5712934538769f803a444ec083540de6b73d901cdcceaae22",
  receiptsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  sha3Uncles: "0x1dcc4de8dec75d7aab85b567b6ccd41ad312451b948a7413f0a142fd40d49347",
  size: 538,
  stateRoot: "0x6907699bd5873eda1b95ee22a4a826b8883a80761eceeebeaa89a9526b6e6ae0",
  timestamp: 1775577083,
  totalDifficulty: 7522001,
  transactions: [],
  transactionsRoot: "0x56e81f171bcc55a6ff8345e692c0f86e5b48e01b996cadc001622fb5e363b421",
  uncles: []
```

Checking the admin node info

```
> admin.nodeInfo
{
  enode: "enode://a3e0ec411b9c6c658245a289eb62ae41f93a67c9839cf01d06840f25ea35bd73417385b2fe1b7c7fc18b1f111b097e6ac0981ace7aee219532c
88296e496b93b@49.43.24.168:30303",
  enr: "enr:-Je4QPY4FL-2hvTHUIROVBWfA5vtBdPiMOX_zviaV0cVVt5hUajzdpP7ib6n-Uh_Ug49_GuSmJbhKBeIlLiMJm6DJ4QOg2V0aMfGhI3ZuOuAgmlkgnY0gmlwh
DErGKiJc2VjcDI1NmsxoQOj4OxBG5xsZYJFoonrYq5B-TpnyYOc8B0GhA8l6jW9c4N0Y3CCdl-DdWRwgnZf",
  id: "eeeee1fa666f7c09061bfc3f965eab385282298806e745cd584c1b3ecdbf4261",
  ip: "49.43.24.168",
  listenAddr: "[::]:30303",
  name: "Geth/v1.9.25-stable-e7872729/windows-amd64/go1.15.6",
  ports: {
    discovery: 30303,
    listener: 30303
  },
  protocols: {
    eth: {
      config: {
        byzantiumBlock: 0,
        chainId: 1234,
        constantinopleBlock: 0,
        eip150Block: 0,
        eip150Hash: "0x0000000000000000000000000000000000000000000000000000000000000000",
        eip155Block: 0,
        eip158Block: 0,
        homesteadBlock: 0,
        istanbulBlock: 0,
        petersburgBlock: 0
      },
      difficulty: 7522001,
      genesis: "0x6f14b168d12c82428b52b485bddd66c77c8daf3c38dcc5c2678d03d260a2805e",
      head: "0xa2f12559d9f4bca9134f88ac1485816862f83f12e4e29d4a168bf0de50bf2cc6",
```

Conclusion:

A private blockchain was successfully implemented using Geth. The blockchain was initialized using a custom genesis block, and accounts were created for transactions. Mining was performed to generate Ether, and a transaction between accounts was successfully executed and verified. This demonstrates the working of a decentralized blockchain network in a controlled environment.